From plan to code: every promise of meeting 1, where it is, and the evidence

Sources: `progressmeet1/future_plan.pdf` (Study guide 2, 16 Oct) and `progressmeet1/slides.pdf` (meeting 1). Status on 24 Oct 2026. “Evidence” is a measurement from a real run on PostgreSQL master `26bd08fb` unless it says “unit test”.

The three claims (slides 6, plan 0.3)

Claim	Status	Where	Evidence
One tree, four jobs: generation, mutation, metamorphic rewrite, reduction on the derivation tree	done, all four	`grammar/engine.py` (spans, `MINIMAL` marker), `grammar/generator.py` (replay), `grammar/rewrite.py`, `triage/tree_reduce.py`, `grammar/treeio.py`	BUG-01 case: 116 tokens (token reduction) vs **79 tokens** (tree then token); 599/600 test cases replay identically in a fresh process (`docs/evidence/tree_reduction/`)
Soundness we measure: generator validity, oracle precision	done	`eval/metrics.py`, `triage/precision.py`, `triage/verdicts.yaml`	validity 99.6-99.7 % on master; precision ledger of 18 triaged candidates, before/after fixing each false-positive class
The LLM proposes; the server decides	done, extended to oracle rewrites	`llm/gate.py`, `llm/loop.py`	live gate: valid fragment accepted, invalid rejected with the server’s own error; rewrite gate accepts identities, rejects `e+1` (12 mismatches) and `e IS TRUE` (value probe)
Localization from plan oracle x rewrite oracle (plan 0.3)	done	`triage/localize.py`	BUG-02 localized to “not toggled by planner settings” = the planner support function, its true cause

Plan item by item

Plan section	Item	Status	Where / evidence
0.1-1, 2	Grammar arm tuned and fuzzing for hours	done	6-hour hunts; 99.7 % valid statements
0.1-2, 3.2	TLP (WHERE, DISTINCT via UNION, HAVING)	done	`rewrite._tlp_where/_tlp_having`; also inside derived tables, CTEs, set arms, IN/EXISTS/LATERAL
3.2	TLP variants for aggregates (MIN/MAX, SUM of SUMs, SUM of COUNTs)	**done (new)**	`rewrite._tlp_agg` (+ `BIT_*`, `BOOL_*`, `EVERY`)
3.3	NoREC	done	`rewrite._norec`
3.4	`x IN (a,b,c)` <-> `x=a OR x=b OR x=c`	**done (new)**	`rewrite._pred` (`pred:in_or`)

Plan section	Item	Status	Where / evidence
	(OR-to-ANY)		
3.4	`p -> p AND c IS NOT NULL` for a NOT NULL column	**done (new)**	`rewrite._notnull` (not under outer joins, not for generated columns)
3.4	CTE `MATERIALIZED` toggle	done	`rewrite._cte`
3.4	`e -> CASE WHEN true THEN e END`, `e -> (SELECT e)`	done	`rewrite._ANY` (found BUG-02)
3.4	`a < b -> NOT (a >= b)`	**done (new)**	`pred:cmp_negate` (exact under 3VL, no guard needed)
3.4	`t -> (SELECT * FROM t)` subquery flattening	done	`table_fence` (`OFFSET 0`)
3.4	ORDER BY removed when compared as a multiset	**done (new)**	`rewrite._order_drop`
3.4	BETWEEN expansion (extra)	**done (new)**	`pred:between`
3.4	“letting the LLM add new rewrites is our contribution”	**done (new)**	prompt `rewrites` field; `gate.check_rewrites` (value probe + 150 live test cases, zero mismatches); `results/llm/rewrites` feed the LLM arms
3.4	`t JOIN s ON k` -> `WHERE EXISTS` for unique `s.k`	not done	needs uniqueness proofs over joins; join removal is exercised by the table fence and `no_self_join_elim` / plan variants instead
3.5	Reuse `eligible()` determinism filter and digests	done	`eval/differential.py`
3.5	At most 1-2 rewrites per query	done	`mm_per_query`
3.5	Re-run both sides before reporting	done	`mm_confirm`
3.5	Re-run under planner variants: planner bug vs executor/rewrite bug	**done (new)**	`triage/localize.py`, in every triage report
4	Cross-version differential (release vs master)	done	`reference/xversion_executor.py` (18.6; REL_19 does not exist yet), per-worker reference DB, intended changes excluded (`corr/regr_*`)
5.1	Commit miner since the branch point	done	`llm/commits.py` (`--exclude` keeps benchmark fixes away from the

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			model)
5.1	LLM proposes grammar edits and seeds	done	`llm/prompt.py`, `llm/client.py` (API or manual mode, stored responses)
5.1	Validation gate: parse, sample, validity, coverage of changed lines	done	`llm/gate.py`
5.2	Diff coverage metric	done	`llm/diffcov.py`, per-worker gcov counters (`scripts/rq1_servers.sh`)
5.3	Feedback-weighted alternatives (bandit)	**done (new)**	`generator._bandit_update`; one run: 533 vs 419 plan shapes at equal validity; ablation arm `grammar_mm_flat` in RQ1
5.4	Acceptance rate, token cost, coverage per patch	done	`pgfuzz llm report` (ledger)
5.5	Hot targets: eager aggregation needs a GUC variant	**done (new)**	plan variant `no_eager_agg` (18 variants now)
5.5	Skip scan, OR-to-ANY, virtual generated columns, RETURNING OLD/NEW, MERGE, JSON_TABLE, partitionwise	in grammar	multi-column indexes, `pred:in_or`, table `t4`, grammar rules
6	Plan-shape novelty feedback (QPG)	done	`plan_shapes` in executor, corpus reason `plan`
6	Boundary literals from interval analysis (L19-20)	**done (new)**	`grammar/profile.py`: observed min/max +-1, type edges (NaN, +-Infinity, -0, extremes, empty array)
6	Data-aware predicates with target selectivity (L27-28)	**done (new)**	`pgfuzz profile` + `@dlit(T)`: predicates selecting 5-95 % of rows **29.3 % -> 58.6 %**
6	Schema churn (ANALYZE, indexes, tables between queries)	done	`setup_stmt` rules (VACUUM cannot run inside the test transaction)
7.1	Reproduce x3, park flaky	done	`triage --repeat 3`
7.2	Grammar-tree reduction, then ddmin on statements and tokens	**done (tree part new)**	`triage/tree_reduce.py` + `triage/reduce.py`

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7.3	Classify crash / assertion / XX000 / wrong result	done	`eval/oracle.py` (+ crashes of other sessions no longer blamed)
7.4	Bisect	done	`scripts/bisect_pg.sh`
7.5	Check the latest release	done	`--reference`, cross-version oracle
7.6	Dedup against pgsql-bugs / hackers archives	manual	checklist in every `report.md` (archives are not reachable from the fuzzing machines)
7.7	Report with a fixed template	**done (new)**	`upstream_report.txt` per candidate
7.8	Precision per oracle (L5)	done	`pgfuzz precision`, before and after fixed classes
8.1	RQ1-RQ4	designed, configs ready	`configs/rq1.yaml` (7 arms), `configs/rq3.yaml` (4 arms)
8.2	Re-injected bug benchmark, 8-12 fixes	**done, 23 fixes (20 observable)**	`bench/reinjected/`, `scripts/build_reinjected.sh`, `pgfuzz bench verify/attribute`
8.3	Medians with 95 % CI, Mann-Whitney U, A12	**done (CI new)**	`eval/stats.py` (`median_ci`), report tables
8.4	Five graded paper parts	drafted	`paper/` (motivating example = BUG-02, a real new bug)
9 (L3, L21-22)	Stretch: RLS leak oracle (information flow)	**done (new)**	`corpus/rls.sql`, kind `rls`: 532 live checks, 0 false positives
0.1-7	Stretch: CERT-style performance oracle	**done (new)**	kind `cert`, outcome `performance`

Beyond the plan (added to raise the level)

Addition	Why it matters
Value probe for LLM rewrites	sampling alone accepted the false identity `e IS TRUE` (109 checks); the probe rejects it at NULL: a measured lesson for the paper (Observation 1)
Foreign-crash filter (PID matching)	a crash caused by another session is no longer blamed on the running test case: found by our own smoke run
Numeric-scale / float `-0` canonical digests	removed a false-positive class that a richer grammar exposes
Pinned base commit for all RQs	RQ1-RQ4 and diff coverage measure the same code
Per-worker gcov counters and reference databases	parallel trials cannot contaminate each other
Upstream report drafts and localization	shortens the path from a candidate to an accepted bug (the 50 % component)

Honest gaps

- `t JOIN s` -> `EXISTS` rewrite (needs uniqueness reasoning; covered indirectly).
- Archive de-duplication is a manual checklist step.
- The paper is over the page limit until the result tables decide what to cut.
- Every number above is a component check; the campaign numbers (RQ1-RQ4) come from `docs/EXECUTION_PLAN.md`.